

SIMATSENGINEERING

ASSESSMENTTOOL3

NAME:M. Jayaprakash
COURSENAME:CSA1607

REG.NO:192425268
COURSECODE:DataWarehousingandDataMining

COURSEOUTCOMECOVERED

CO4:Examinetheapplicabilityofclusteringalgorithmsinreal-worldscenarios,presenting findings through clear reports with effective visualizations.
Assessment ToolWeightage:20%

QUESTIONS:05

TotalMarks:50

AnnexureA

COMPARITIVEANALYSIS

Criteria	Scopeof Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking(2)	Clarity of Presentation(1)	Total(10)
Weightage	20%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

CodeofConduct:

*M. Jayaprakash(192425268)(Name/RegNo)certifythatthissubmissionismyoriginalwork and that I have adhered to the guidelines specified for this assessment.
Iunderstandthatanyviolationofacademicintegrityruleswillresult indisiplinaryaction.*

Signatureofthestudent:
jayaprakash

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

AnnexureA

Q1.Dataset:AssessmentTool3:ComparativeAnalysis(25%→5 Marks)

Timeduration:45Minutes

AssessmentTool3 (Low)

Q1.Dataset: CustomerData

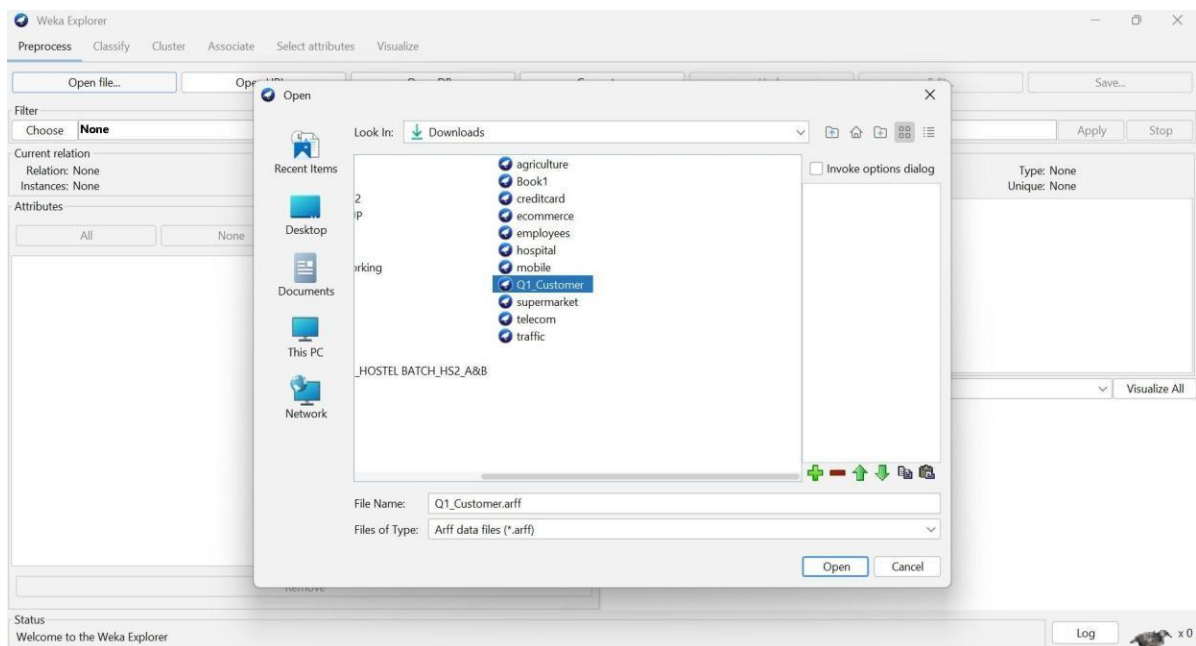
CustomerIncome(₹)SpendingScore

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

CompareK-meansandhierarchicalclusteringusingthisdatasetandexplaindifferencesin performance and scalability. (BL3 – 1 Mark)

OUTPUT:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

(Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalClusterer

Clusterer output

Initial starting points (random):

Cluster 0: C4,28000,65

Cluster 1: C2,60000,40

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
	(5.0)	(3.0)	(2.0)	
Customer	C1	C1	C2	
Income	46600	36000	62500	
SpendingScore	54	65	37.5	

Time taken to build model (full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	3 (60%)
1	2 (40%)

Status OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

(Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalClusterer

Weka Clusterer Visualize: 06:38:09 - SimpleKMeans (customer_clustering)

X: Instance_number (Num)

Y: Customer (Nom)

Colour: Cluster (Nom)

Reset Clear Open Save

Jitter

Plot: customer_clustering_clustered

Class colour

cluster0 cluster1

Status OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file...

Open

Look In: Downloads

Recent Items

2

p

Desktop

Documents

Working

This PC

Network

Invoke options dialog

Type: None

Unique: None

Visualize All

File Name: Q1_Customer.arff

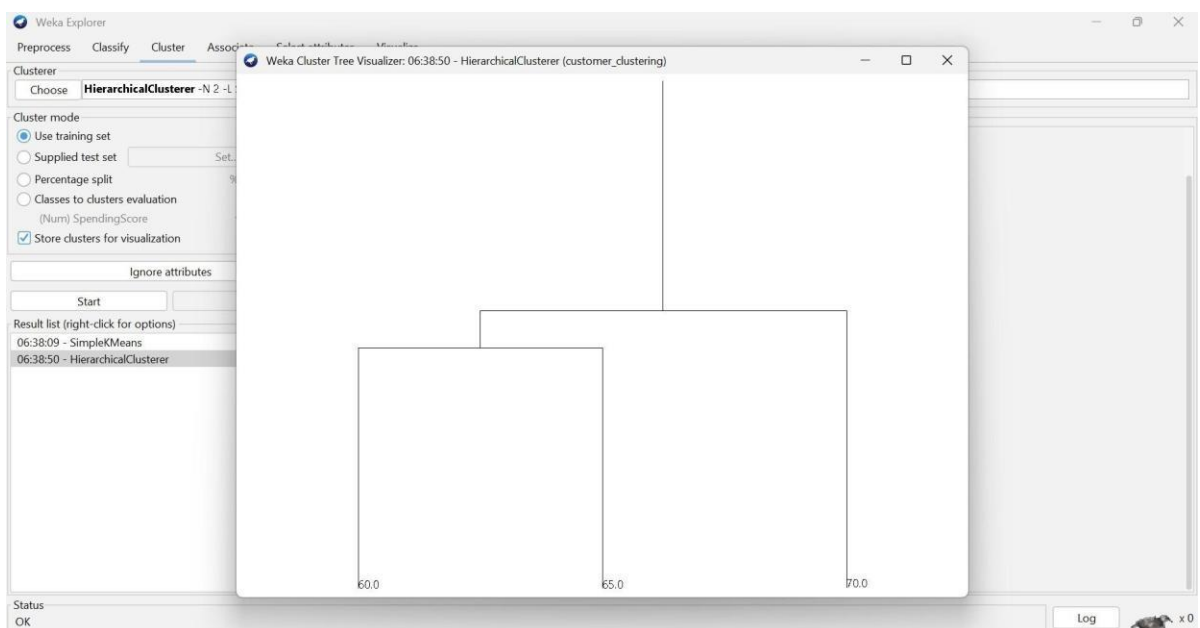
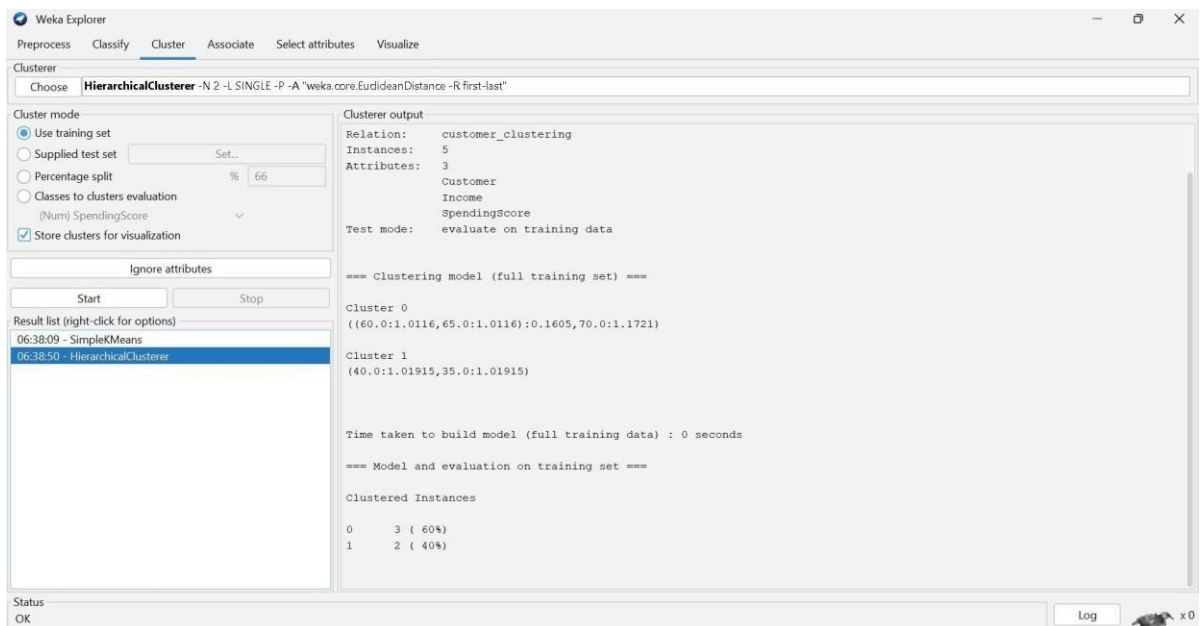
Files of Type: Arff data files (*.arff)

Open Cancel

Status

Welcome to the Weka Explorer

Log



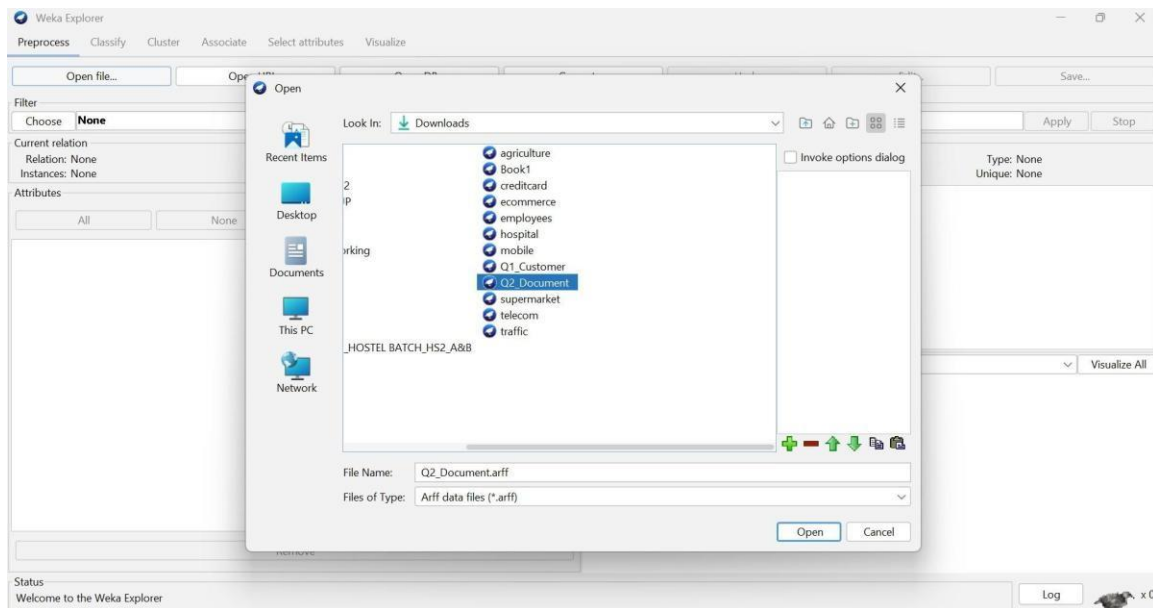
Q2.Dataset:DocumentData

DocumentWord1Word2Word3

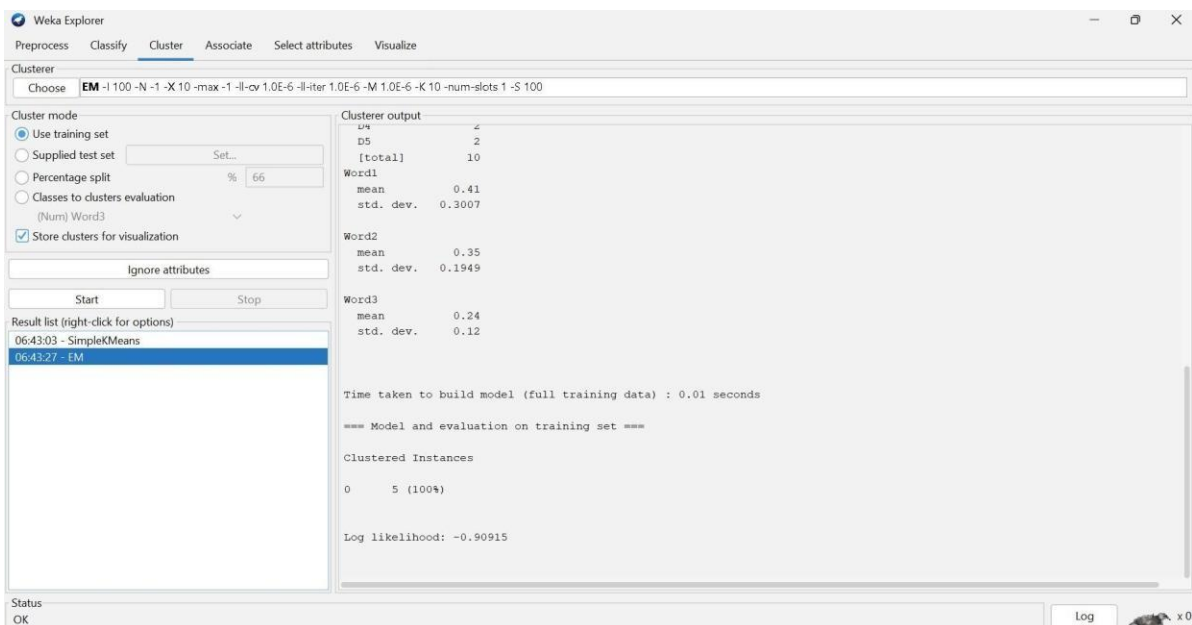
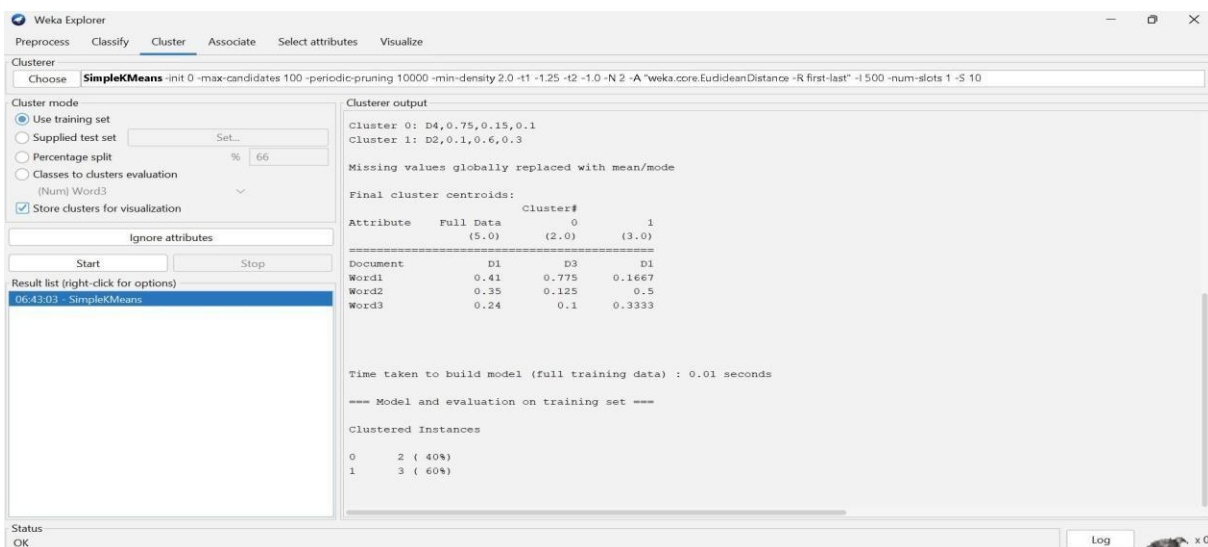
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

CompareK-meansandEMalgorithmsinhandlingoverlappingclustersfor thegivedataset.
(BL4–1 Mark)



OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans

Cluster output

Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
			(5.0)	(2.0) (3.0)
Document	D1	D3	D1	
Word1	0.41	0.775	0.1667	
Word2	0.35	0.125	0.5	
Word3	0.24	0.1	0.3333	

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

Status OK Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans
06:43:27 - EM

Cluster output

D5	2
[total]	10
Word1	
mean	0.41
std. dev.	0.3007
Word2	
mean	0.35
std. dev.	0.1949
Word3	
mean	0.24
std. dev.	0.12

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	5 (100%)
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Log likelihood: -0.90915

Status OK Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans
06:43:27 - EM

Weka Clusterer Visualize: 06:43:27 - EM (document_clustering)

X: Instance_number (Num) Y: Document (Nom)

Colour: Cluster (Nom) Select Instance

Reset Clear Open Save Jitter

Plot: document_clustering_clustered

Class colour cluster0

Status OK Log

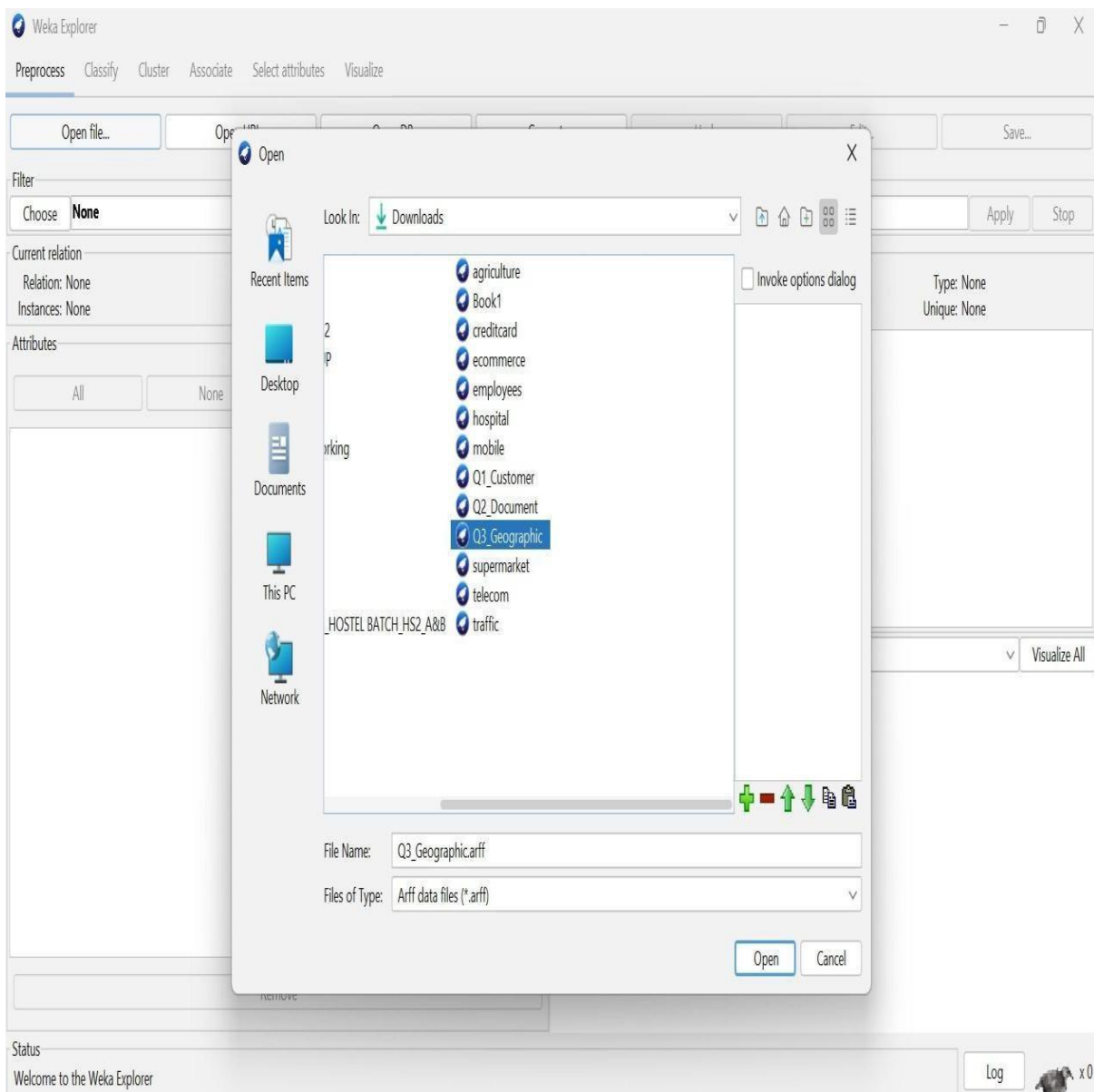
Q3.Dataset:Geographic Data

PointLatitudeLongitude

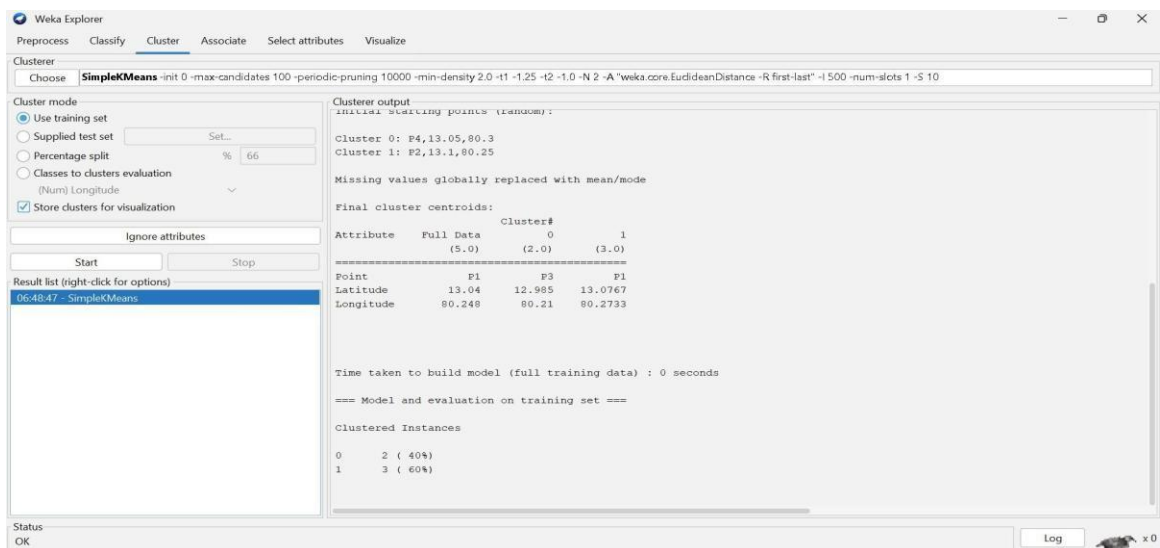
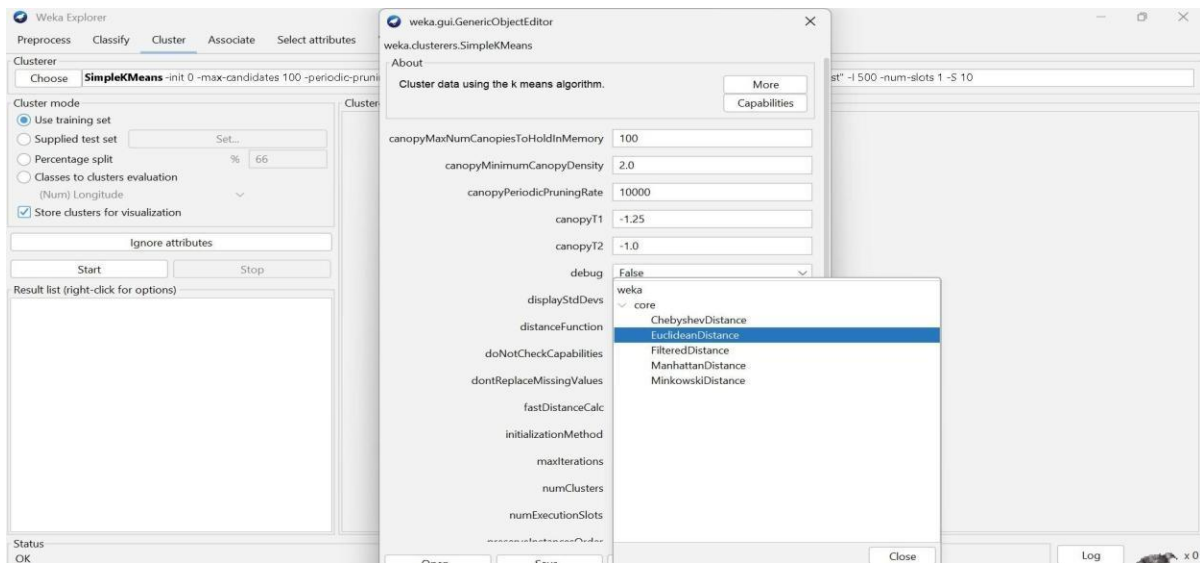
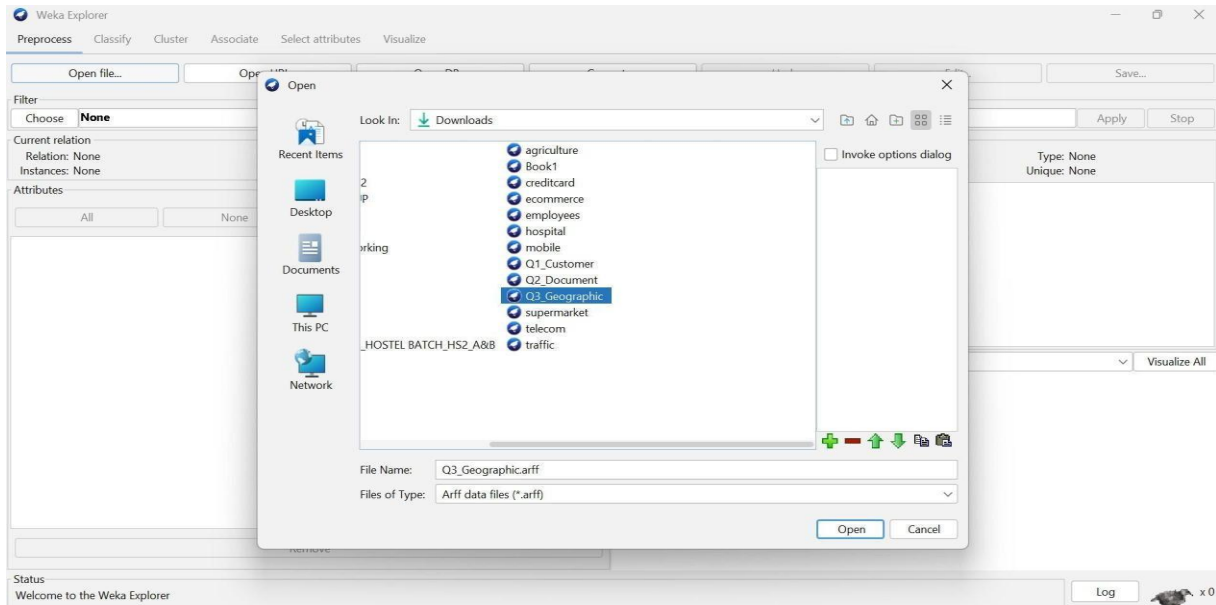
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

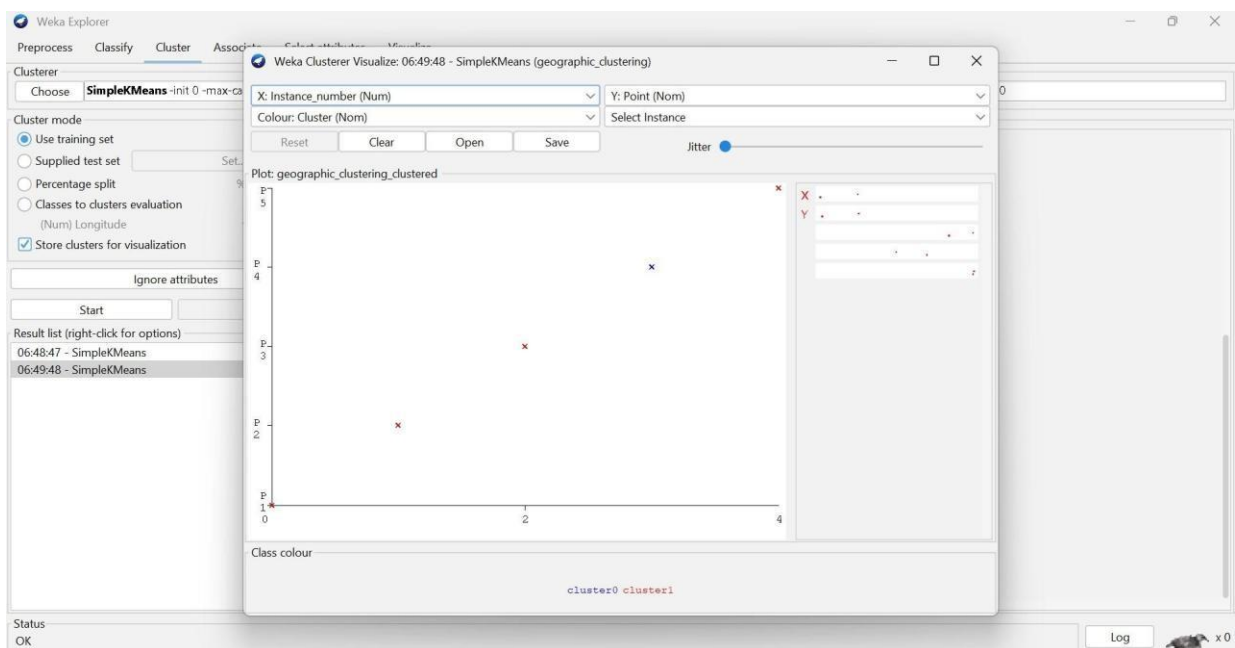
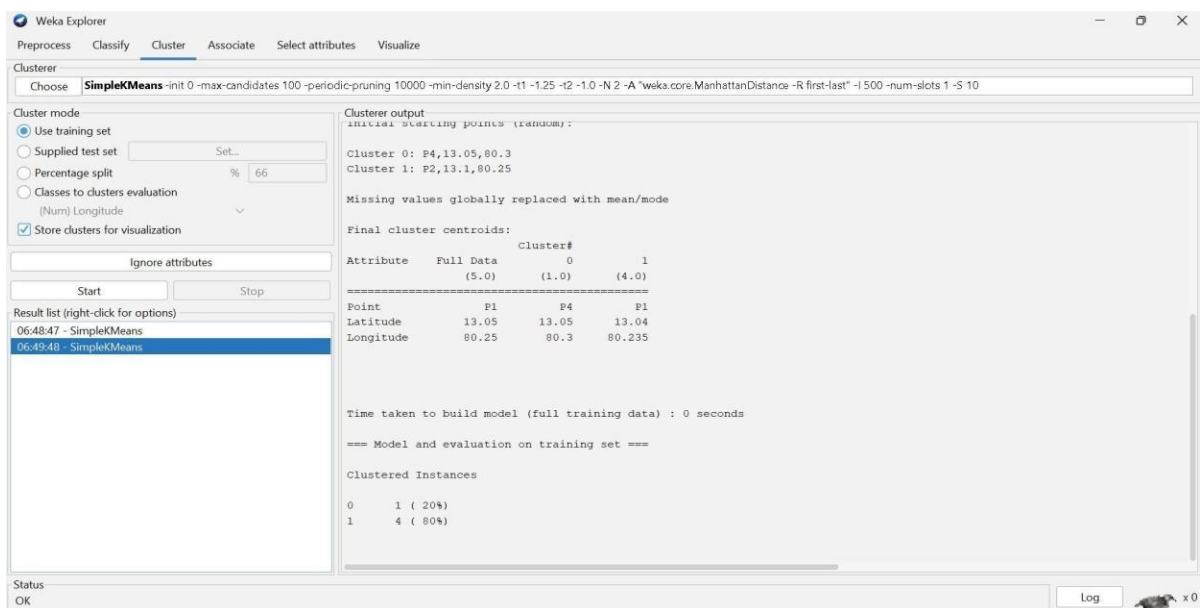
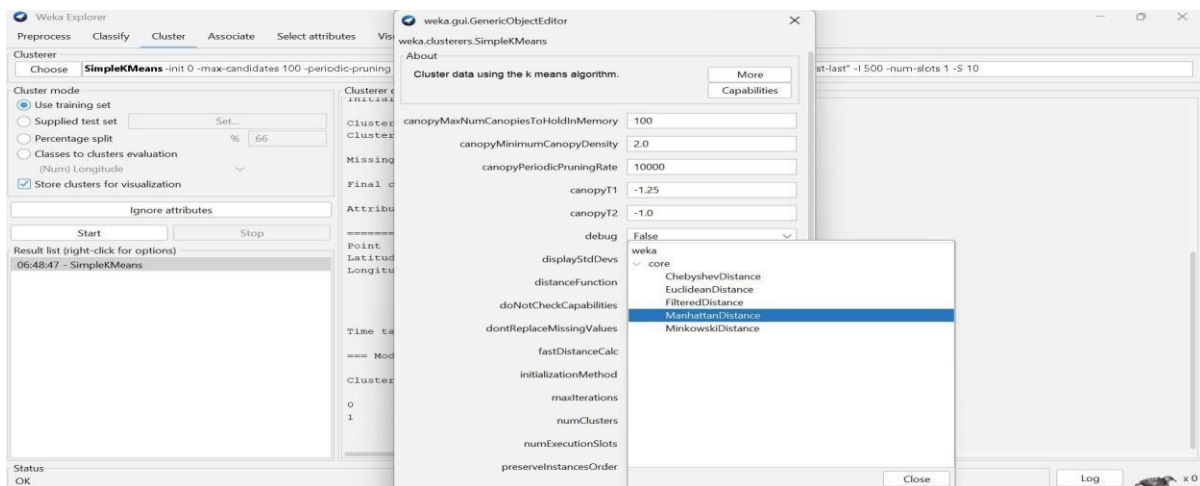
Question:

CompareEuclideanandManhattandistancemetricsandanalyzetheirimpactonclustering results.
(BL4 – 1 Mark)



OUTPUT:





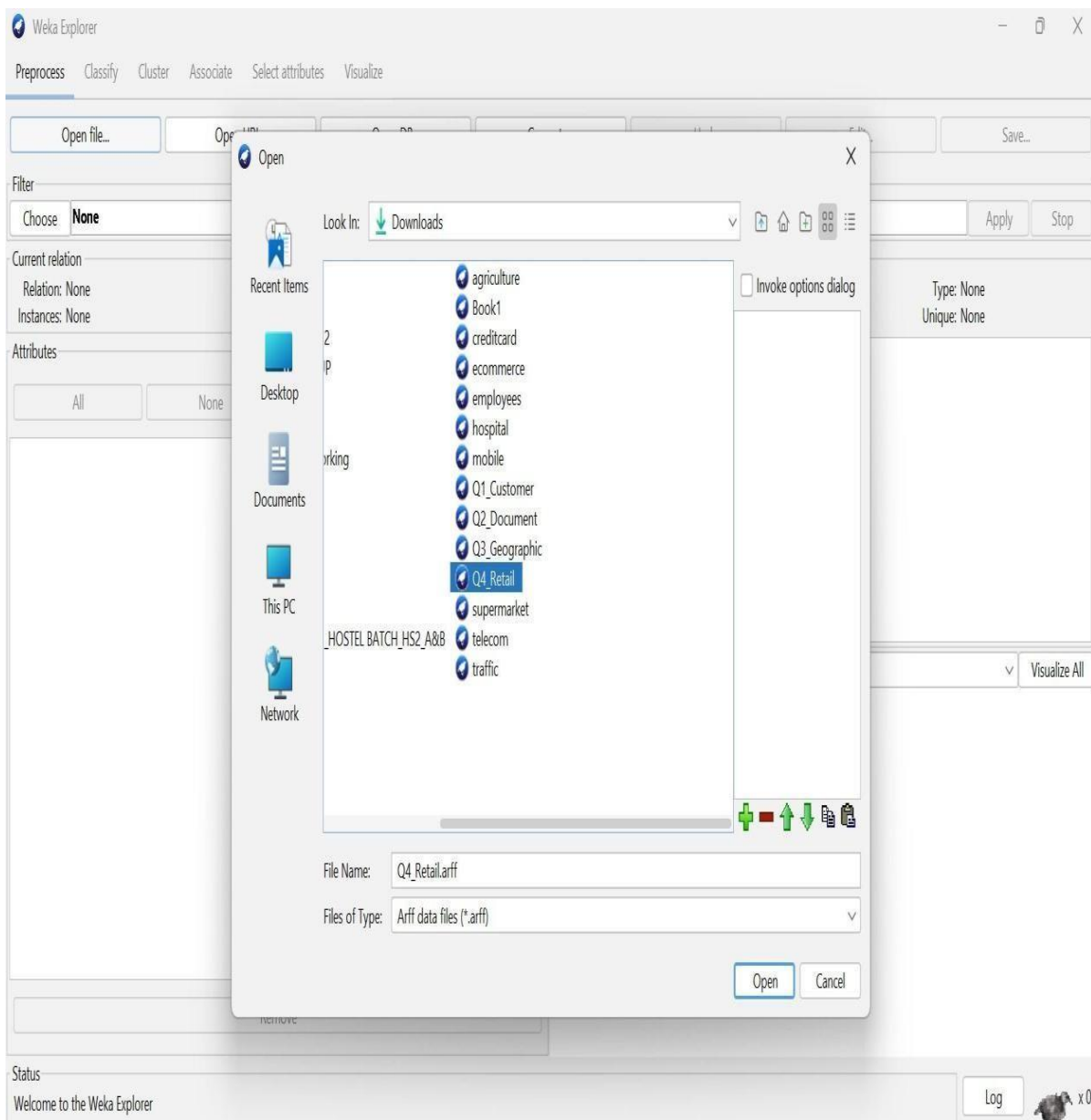
Q4.Dataset:RetailProducts

ProductSales(₹)Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)



OUTPUT:

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'HierarchicalClusterer' is chosen, and the 'Use training set' option is selected under 'Cluster mode'. The 'Clusterer output' pane displays the following information:

Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

The 'Result list' on the left shows the job '06:53:39 - HierarchicalClusterer' as the most recent entry. The status bar at the bottom indicates 'OK'.

Weka Explorer

Preprocess **Classify** **Cluster** Associate Select attributes Visualize

Clusterer: Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split %
- ☐ Classes to clusters evaluation (Num) Frequency
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options):

- 06:53:39 - HierarchicalClusterer
- 06:54:50 - MakeDensityBasedClusterer**

Cluster output:

```

Discrete Estimator. Counts = 1 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

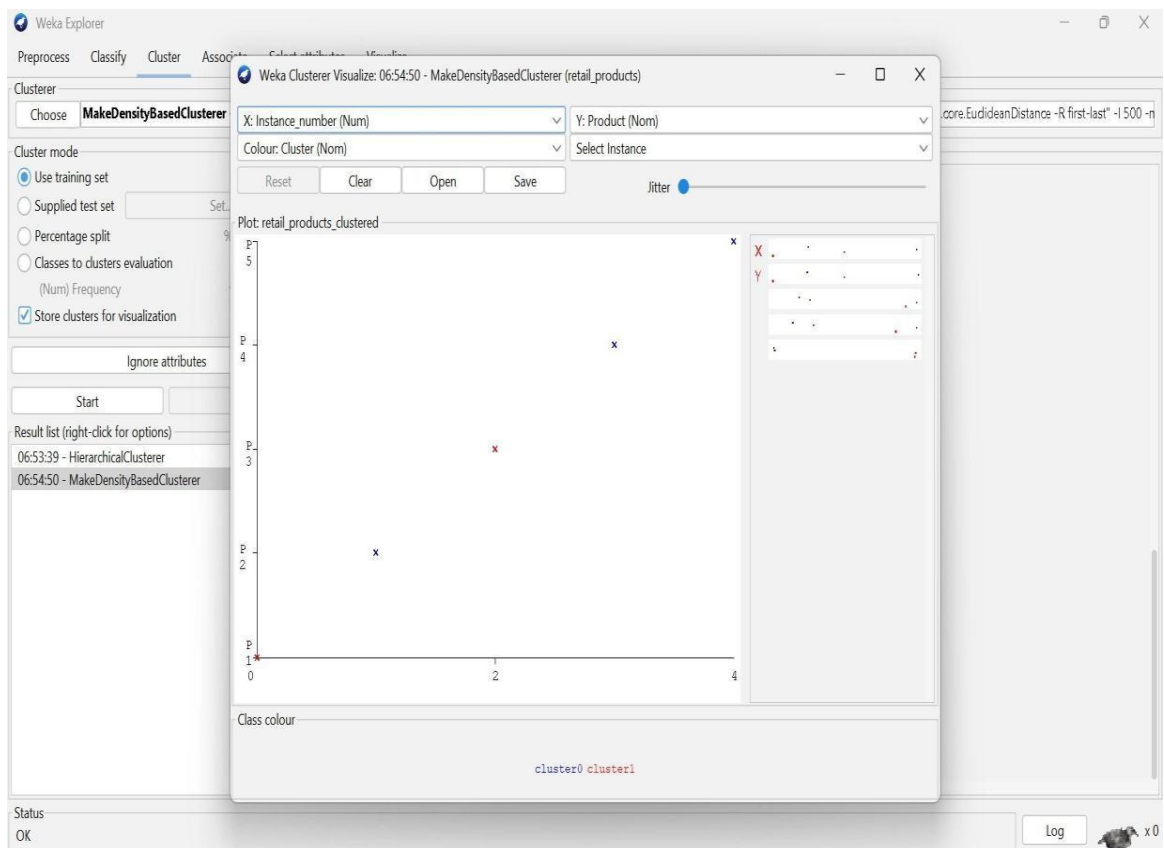
Clustered Instances

0      3 ( 60%)
1      2 ( 40%)

Log likelihood: -12.6953

```

Status: OK x 0



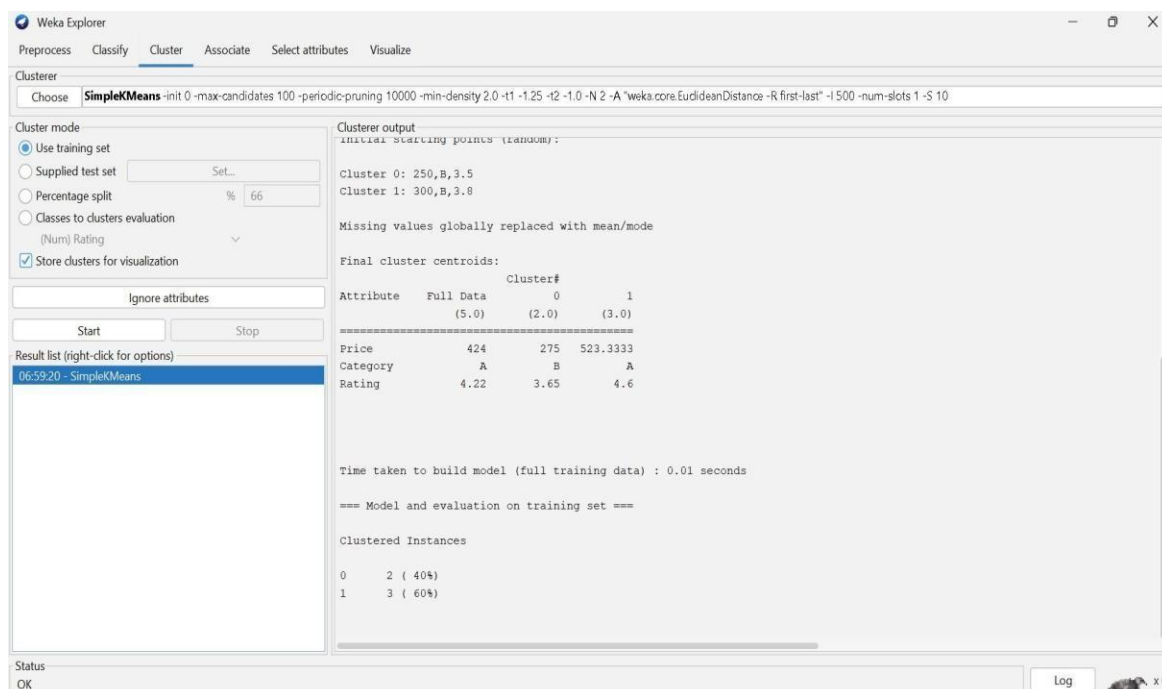
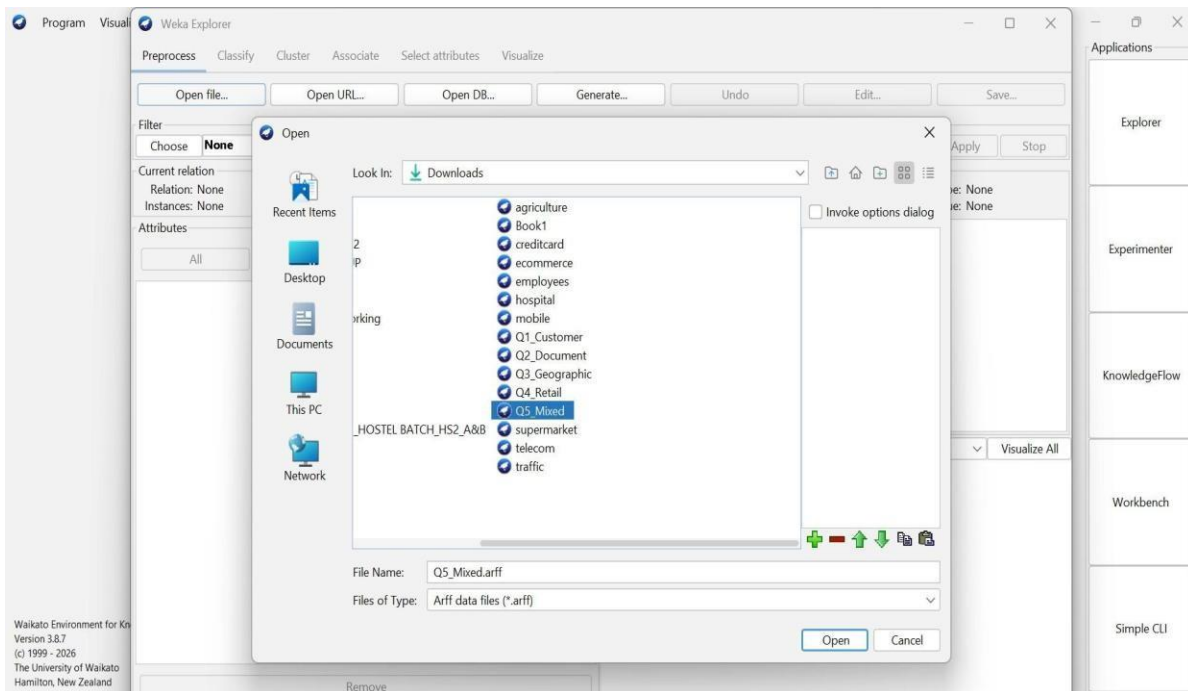
Q5.Dataset:MixedAttributes

ItemPrice(₹)CategoryRating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)



OUTPUT:

